Intelligent Learning Models for Cancer Diagnosis and Treatment: A Comprehensive Analysis

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***Abstract*— Cancer poses a significant global health and economic challenge, with 20 million new cases and 9.7 million deaths reported in 2022 itself. Early detection of cancer is crucial for improved survival rates, reduced mortality, making it imperative to develop advanced diagnostic tools. Intelligent Learning Models (ILMs) utilizing Deep Learning (DL) and Machine Learning (ML) by leveraging large datasets and complex pattern recognition are posing as the potential diagnostics solution for cancers. This paper reviews the current state of ILMs, evaluates existing models and case study for diagnosis prognosis and treatment of cancers.**

Keywords*— *machine learning, deep learning, intelligent learning models, prediction, cancer.

I. INTRODUCTION

Across the globe in the 21st century, cancer poses a significant challenge for society, public health, and the economy. According to recent data from the International Agency for Research on Cancer (IARC), part of the World Health Organization (WHO), a staggering 20 million new cancer cases were diagnosed in 2022, tragically leading to 9.7 million deaths worldwide.

A concerning statistic revealed in the report is that roughly one in five people will develop cancer at some point in their lives. Additionally, the disease claims the lives of approximately 1 in 9 men and 1 in 12 women [1]. A closer examination of the data reveals that cancer contributes to approximately one in six fatalities (16.8%) worldwide and one in four (22.8%) from noncommunicable diseases (NCDs). Figure 1 and Figure 2 shows the number of age standardized incidence and mortality for various cancers worldwide.

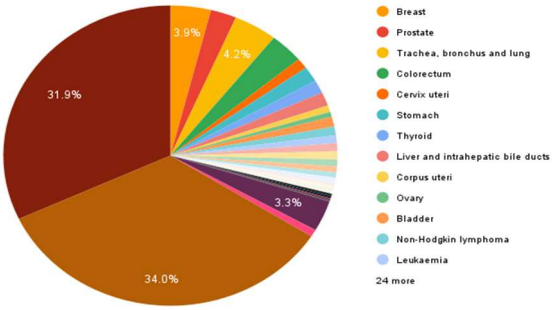


Fig. 1. Incidence of total cancer

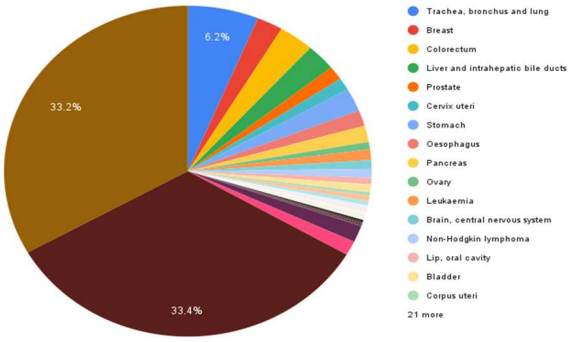


Fig. 2. Mortality of total cancer

Following its onset in the body, cancer progresses through several stages, ultimately culminating in a terminal phase where treatment options are severely limited, the prognosis is poor, and the disease becomes incurable and fatal. Therefore, early detection of cancer is crucial for improving survival rates and reducing mortality among affected individuals. Various researchers and organizations are developing Intelligent Learning Models (ILMs) to cater the diagnostics challenges present in the management and treatment of the cancer. Despite of the advantages, AI (Artificial Intelligence) also comes with some challenges which includes, acceptability of AI based healthcare systems, enhancement in payment structures, data annotation and curation for AI model training and lack of standardized EHRs. Furthermore, it is difficult to obtain a large dataset gathered in a clinical setting, which is required for AI model training. Apart from these, lack of efficient feature-selection methods and AI model generalizability across different regions there are some technical challenges which are inhibiting the AI adoption in medical systems [37].

There's been a growing interest in using AI and ILMs to analyze electronic health records (EHRs) and medical images for cancer diagnosis [1]. These ILMs, which often combine DL and ML, can automatically learn and identify complex patterns from large datasets using various statistical and neural network models [2]. This study explores existing diagnostic

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solutions that leverage AI and ILMs for different cancer types. The objective of this study include:

- Examining the recent studies involving ILMs and cancer.
- Evaluate and compare existing models for diagnosing, prognosing, and treatment of cancer.
- Highlighting the future direction and research focuses on clinical validation, real-time implementation, and multidisciplinary collaborations of AI-driven ILMs for the diagnosis, prognosis, and treatment of cancers.

The paper's structure is as follows: Section II reviews existing literature, followed by Section III's discussion of various cancers and current diagnostic procedures. Section IV provides an overview of intelligent models, while Section V delves into ethical and legal considerations. A case study is presented in Section VI, concluding with a plan to address and overcome obstacles in future research outlined in the final section.

II. LITERATURE REVIEW

AI is being used to reinvent healthcare by researchers in improving the current way of disease diagnosis such as skin cancer and breast cancer and many more. This is very important as most of these diseases are invisible to the naked eye. Research has been carried out in the detection of skin cancer and it is proven that AI models formed in a specific manner called Complex Neural Networks can identify skin cancer with the highest accuracy. There is even one system that was over 99-percent accurate at identifying various kinds of skin cancer in one study. This is an exciting development, as it may help doctors to detect skin cancer faster, and early detection means better treatment results [3].

AI is also being trained to read mammograms, X-ray images of the breast, to help catch breast cancer sooner. Microcalcifications, which are very small clusters of calcium deposits that might be associated with cancer, have been the subject of studies suggesting that AI could be utilized to identify them. Some other studies tried to apply CNNs to detect and classify breast abnormalities in mammograms. Studies thus far indicate that AI can ultimately lead to more accurate mammograms and fewer false positives [4].

Artificial intelligence appears to be coming to the aid of doctors in battling cancer. One of the most effective AI techniques, deep learning, is currently being harnessed to analyze medical images and enhance the detection of cancer. For example, researchers like Wei et al. identified early stomach cancer via deep learning in stomach endoscopy (AI generates cancer “screening” macros) Although it is just work at the outset, it is possible to make these diagnoses more accurate and reproducible [5].

Cervical Cancer Some other areas using deep learning in this case, researchers have created an algorithm that can identify different types of cervical cancer from images of tissue samples. This work, using ResNet50, has demonstrated high accuracies and could help pathologists in cancer diagnosis [6]. Advancements in AI are also coming to the aid of the fight against breast cancer. Almutairi et al. created a mix of deep learning and another AI practice, reinforcement learning, to present another technique The hope is that this will further complement the identification and classification of breast cancer features in a bid for more precise diagnoses These investigations illustrate the promising capabilities of AI to transform cancer diagnosis and therapy [7].

III. TYPES OF CANCERS AND EXISTING DIAGNOSTICS PROCEDURE

Depending on the site in the body of their onset, cancers are categorized into various types as shown in figure 3. In literature, n number of cancers exist that affect various body organs and lead to the worst quality of life, the five most common vital organs affected by cancers these days include breast, lungs, colorectal, prostate and stomach.

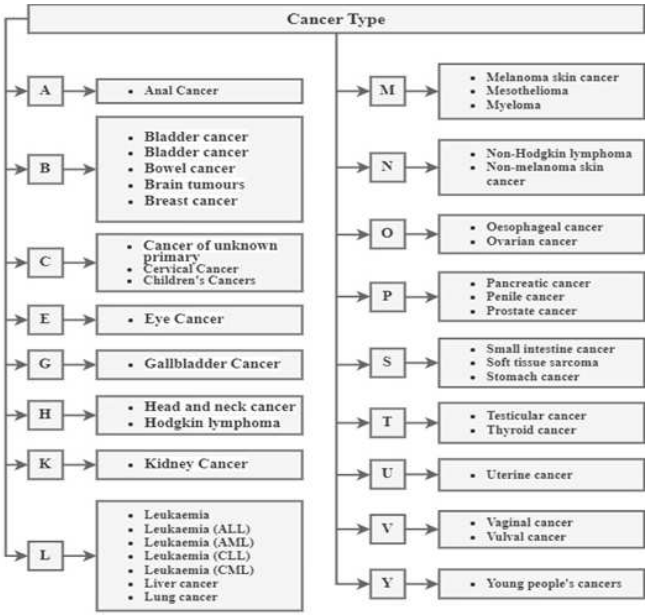


Fig. 3. Types of Cancers Based on their Onset on Body

A. Breast cancer

This is one of the most widespread forms of female cancers worldwide. The disease is characterized by various histological types (e.g., ductal, and lobular carcinoma), hormone receptor status (ER, PR positive or negative), HER2 status and genetic mutations (e.g., BRCA1/2). The traditional methods to detect these cancers involve mammography, ultrasound, MRI, and biopsy [8].

B. Lung Cancer

There are two main categories of lung cancer: non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC). They both basically incur due to smoking or environmental factors and cause persistent cough, shortness of breath, and chest pain in the diseased person. These cancers are typically detected by imaging tests and biopsies [9].

C. Colorectal Cancer

These cancers often start in the colon or rectum and are influenced by factors such as lifestyle, age, family history, and genetic mutations. Changes in bowel habits, rectal bleeding, and stomach pain are among the symptoms that adversely affects the quality life of diseased persons. The screening techniques used in diagnosis of these cancers include stool tests and colonoscopies [10].

D. Prostate cancer

This is also called adenocarcinoma which originates in the prostate gland and causes problems like sexual dysfunction, pelvic pain, and urine problems. These cancers are usually

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impacted by lifestyle, genetics, and age. The diagnosis of these cancers involves digital rectal exams, biopsies, imaging, and PSA testing. An early-stage malignancies typically have a good prognosis for these cancers however it also depends on the individual and disease progression [11].

E. Stomach (gastric) cancer

This primarily arrives as adenocarcinoma and often shows no symptoms at the beginning, however as it progresses symptoms like stomach pain, weight loss and difficulty in swallowing start manifesting into the patients. Usually, smoking, dietary habits, genetic predispositions and Helicobacter pylori infection are the key risk factors associated with it. The common diagnostic procedure for these types of cancers includes imaging, endoscopy, and biopsies. An early identification of these cancers is crucial as advanced stages usually result with worse outcomes during the treatment process [12].

IV. INTELLIGENT MODELS

Intelligent Learning Models comprising ML and DL models are offering various services in cancer prediction, diagnosis, and treatment. Table 1 summarizes research using ML for cancer detection and prediction. Each row details a study, specifying the cancer type, methodology (including feature extraction, selection, classification/prediction tasks), models used, datasets, performance metrics, and the specific task addressed (e.g., classification, detection). DL models and various machine learning algorithms are prevalent, aiming for high accuracy in cancer diagnosis.

TABLE I. MACHINE LEARNING MODELS FOR CANCER DETECTION AND PREDICTION

Source	Cancer type	Task	Model used	Data Types	Performance metrices
[13]	Breast Cancer	-Feature extraction -Feature selection - Classification	-ResNet18 -PSO, ASO, EO -SVM, KNN, DT	Histopathological images	Accuracy-0.9773 F1-score-0.9775
[14]	Breast Cancer	-Feature selection - Classification	-Gorilla Troops Optimization Deep Q-Learning, Deep Neural - Network (DNN)	Mammogram images	Accuracy High
[15]	Cervical Cancer	- Classification -Feature selection -Causal analysis	-SVM, RF, DT -Boruta analysis	Cervical cancer images	Classification Precision-0.8456 Recall- 0.812, F1 Score-0.684 Boruta analysis Precision-0.912 Recall- 0.891 F1 score-0.798
[16]	Invasive Cervical Cancer	Classification -Feature selection techniques	-MLP, LR, RF, kNN, DT, SVM, GB, ADB -Chi-square, SelectBest, RF	Not specified	Accuracy-0.981
[17]	Cervical Cancer	Prediction	DT, SVM	UCI Cervical cancer text dataset	Accuracy-0.9744 Sensitivity-0.9779 Specificity-98.01
[18]	Early gastric Cancer (EGC)	- Classification - Segmentation	-DL, CNNs, Weakly supervised learning -Guided attention mechanism -U-Net based segmentation	Endoscopic images including C-WLI, ME-NBI, NBI, and CE	Accuracy- 0.98 Intersection over Union (IOU) ≥0.64
[19]	Breast Cancer	- Classification -Feature selection	-DT, Deep learning models (VGG16) -Firefly algorithm	Breast thermal images	Accuracy-0.955
[20]	Breast Cancer	- Classification -Feature transformation	-ResNet18, Weighted kNN -Metaheuristic algorithms PSO, DFOA, CSOA	Mammography images	Good detection rates (GDR)-High

Table 2 showcases various DL applications in cancer diagnosis. It details the cancer type, methodology (often involving CNNs for image analysis), specific deep learning models used (e.g., ResNet, Transformer), utilized datasets (e.g., medical images, genetic data), and the achieved performance in tasks like classification and detection.

TABLE II. DEEP LEARNING MODELS FOR CANCER DETECTION AND PREDICTION

Source	Cancer type	Methodology	Model used	Data Types	Performance metrics
[21]	Primary liver cancers	Classification	Weakly supervised learning, DL, ResNet18, CNN	biopsy samples, tumour/non-tumour areas in whole-slide images	--
[22]	Lung Cancer	Classification	Bi-LSTM NN RNN, BERT, DNN	CT, PET/CT reports	--
[23]	Prostate cancer	Classification Detection and Segmentation	-DAIClass CNN -3DAISeg CNN	CT scans	--
[24]	Breast bladder, lung, kidney, melanoma	Classification and detection	ResNet CNN Vision Transformer Transfo Faster R-CNN: RPN-CNN(R-CNN) YOLOv5: CNN	High-resolution fluorescence images of stained cells	Accuracy 0.808
[25]	Skin cancer	Classification	Double AMIS-ensemble model, Modified CNN architectures (EfficientNetV2S, EfficientNetV2B3, ConvNeXtSmall) Combining U-Net, RP-Net, and EfficientNet architectures	Dataset malignant vs. benign	--

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[26]	Breast cancer	Classification Instance selection	-Two-stage deep Multiple Instance Learning -Swin Transformer	Whole Slide Image (WSI) of breast cancer pathology dataset	--
[27]	Non-Small Cell Lung Cancer	Classification	DL, CD-CNN framework	LC25000, The Cancer Genome Atlas (TCGA), and Clinical Proteomic Tumor Analysis Consortium (CPTAC)	--
[28]	Liver Cancer	-Classification -Feature selection	End-to-end deep learning model: CNN	Large dataset of MRI scans and clinical data	--
[29]	Skin Cancer	Detection and classification	CNNs	Dermoscopic images	Accuracy-0.9318
[30]	Lung Cancer	Detection and classification	Multimodal deep learning - CNN	Medical images, genomic data, and clinical records	Accuracy-0.925
[31]	Lung Cancer	Classification Gene selection	-Enhanced Convolutional Neural Network -Improved Whale Optimization Algorithm, -min-max normalization	Gene expression data	Accuracy-0.93
[32]	Breast cancer	Detection and classification	-BL, Mask R-CNN: R-CNN -ResNet50: CNN	Two datasets of breast MRI scans	Accuracy-0.977
[33]	Cervical cancer	Classification classification)	Deep Learning System combining ResNet50	Pathological sections,WSI	Accuracy-0.910
[34]	Prostate Cancer	Classification	Deep learning with ResNet-18	MRI scans	Accuracy-0.945
[35]	Breast Cancer	Classification Feature extraction	-Hybrid Deep Learning CNN-LSTM -Dual modality approach	Mammogram and ultrasound images	Accuracy-0.993

V. Ethical and Legal considerations

When incorporating automated models into cancer diagnosis, there arise some ethical issues that must be very carefully considered. One of the major concerns is, a complete reliance on machine-made diagnoses without any human intervention could lead to erroneous results and thus may invite faulty treatment procedure. Besides this, there are substantial issues concerning patient privacy and data security as ILMs often deal with sensitive health-related information which should remain secure and confidential at every stage. Thus, before using these technologies, their ethical implications should be thoroughly investigated because patient autonomy and rights are more important than the models' technological capabilities.

VI. Case study

To predict and diagnose breast cancer, a work presented in [36] examined the effectiveness of a wide range of machine learning (ML) and deep learning (DL) models. The research team employed a meticulous, step-by-step approach, leveraging data from 569 individuals diagnosed with malignant breast cancer in the Wisconsin Diagnostic Breast Cancer dataset. First, the dataset was subjected to rigorous preprocessing and attribute selection for improved quality of data. The data underwent a meticulous splitting process, with 80% designated for training the models and 20% reserved for testing their performance. To achieve optimal results, a variety of ML and DL algorithms were employed, including k-Nearest Neighbors (kNN), Gradient Boosting (GB), Support Vector Machines (SVM), Neural Networks, rule induction with CN2, Naive Bayes, Stochastic Gradient Descent (SGD), tree models, Multilayer Perceptrons (MLP), and Neural Decision Forests. The training and evaluation of these models were conducted collaboratively using the orange data mining tool and the Python software framework. Ensemble techniques such as bagging and boosting were also employed to check for any improvement in the prediction performance. Models are evaluated using the metrices likes F1-Score, accuracy, precision and recall among others. The study found that the models CN2 rule inducer and GB perform well in terms of sensitivity, however there is no match of MLP when it comes to accuracy and AUC-ROC. Also, MLP performs better in precision, recall as well as F1-Score. Thus, it may be concluded that MLP shows higher predictive capabilities for this data than others and thus it can potentially be used for prediction and diagnosis of cancer in clinical settings. Based on these studies, it can be inferred that there is a need to explore other AI techniques for ILMs. For instance, soft computing techniques [38], such as fuzzy logic [39], genetic algorithms [40], particle swarm optimization [41], and others, could be investigated for further advancements.

VII. Conclusion

ILMs have demonstrated immense potential in transforming cancer care, offering more accurate diagnostics, personalized treatment plans, and improved patient outcomes. In this paper we have comprehensively reviewed and explored the significant advancements and applications of ILMs in the domain of cancer detection and prediction. We have found that various ILMs, specifically DL models, have shown remarkable success in analyzing complex medical data, identifying patterns, and making precise predictions. Thus, we can say that despite various challenges, the ILMs can be used in cancer detection and prediction.

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